

Surface tension correlations from symbolic regression

$$T_r = \frac{T}{T_c}, \quad F_1 = 1 - T_r$$

Equation 1

Complexity: 1

Loss: 19.9917

$$\sigma = 7.611$$

Equation 2

Complexity: 3

Loss: 0.852208

$$\sigma = (1 - T_r)48.25$$

Equation 3

Complexity: 5

Loss: 0.00143133

$$\sigma = ((1 - T_r)26.19)^{1.351}$$

Equation 4

Complexity: 7

Loss: 0.000107575

$$\sigma = (((1 - T_r)29.08) - 0.1882)^{1.297}$$

Equation 5

Complexity: 9

Loss: 1.35766e-5

$$\sigma = \left((((1 - T_r))^{0.191} 2.557) - 0.3853 \right)^{5.578}$$

Equation 6

Complexity: 11

Loss: 1.34996e-6

$$\sigma = \left((((1 - T_r))^{0.3445} 4.148) - 0.4036 \right)^{3.268} + 0.1053$$

Equation 7

Complexity: 13

Loss: 3.61994e-7

$$\sigma = \left(((1 - T_r) - 0.6032) + (((1 - T_r))^{0.4465} 4.589) - 0.06824 \right)^{3.128}$$

Equation 8

Complexity: 15

Loss: 1.04279e-7

$$\sigma = \left((((1 - T_r) - 0.369) - 0.2075) + (((1 - T_r))^{0.4333} 4.839) \right)^{2.96} + 0.05994$$

Equation 9

Complexity: 19

Loss: 7.34542e-8

$$\sigma = \left((((1 - T_r) + ((1 - T_r)0.05491)) - 0.3683) + (((1 - T_r))^{0.4356} 4.841)) - 0.1968 \right)^{2.962} + 0.05516$$